

Mohamed Baha Riahi

+216 52 817 709 | ✉ mohamedbaha.riahi@supcom.tn | 🌐 mohamedbahariahi | 📱 BahaRiahi3

EDUCATION

Higher School of Communication of Tunis (Sup'Com)	Ariana, Tunisia
<i>Engineering Degree in Information and Communication Technologies (ICT)</i>	Sep 2024 - Expected Jun 2027
• Relevant Courses: Deep Learning, Cybersecurity, Web Development, Data Structures & Algorithms	
Tunis Preparatory Engineering Institute (IPEIT)	Tunis, Tunisia
<i>Preparatory Classes for National Engineering Exams (Math-Physics)</i>	Sep 2022 - Jun 2024
• Relevant Courses: Calculus, Linear Algebra, Algorithmics & Programming, Database Systems	

EXPERIENCE

Cybersecurity Intern	Tunis, Tunisia
<i>SFM Technologies</i>	Jul 2025 - Aug 2025
- Deployed & integrated simulated SSH services to capture brute-force attacks, configured port forwarding via pfSense to route all port 22 traffic exclusively to the Cowrie SSH honeypot for isolated threat monitoring. - Built a segmented network honeypot architecture by designing and implementing a research-paper-inspired topology in GNS3, isolating the honeypot and sensor from the main Windows Server and Active Directory environment using pfSense firewall rules. - Operationalized log monitoring & alerting by configuring Cowrie as a Wazuh agent, with a central Wazuh manager on a dedicated sensor VM, enabling centralized log aggregation, dashboard visualization, and real-time attack detection. - Performed adversarial simulation & validation by conducting brute-force simulations (Hydra) and reconnaissance scans (Nmap) against the honeypot, analyzed generated logs and built initial visualizations, including API-based log enrichment using Google Gemini.	

PROJECTS

Deep Learning-Based Malware Detection System <i>PyTorch, CUDA, scikit-learn</i>	Jan 2026 – Present
- Built and deployed four complementary architectures - Bidirectional LSTM with attention, Transformer encoder, ResNet-style CNN (3-channel 10×10 image representation), and Graph Attention Network (GAT) - each leveraging unique feature engineering approaches. - Achieved 98% accuracy and 0.99 ROC-AUC on imbalanced dataset (39.6:1 malware ratio) using weighted sampling and class-weighted loss functions. - Created FastAPI microservice with ensemble prediction, Docker containerization, Nginx load balancing, and achieved less than 100ms inference time, Providing real-time behavioral malware detection without signature dependencies, suitable for endpoint security and sandbox integration.	
Athena AI (LLM Study Assistant) <i>FastAPI, ChromaDB, Ollama</i>	Dec 2025 – Jan 2026
- Built an intelligent RAG-based study companion, AthenaAI, a full-stack application enabling users to upload PDFs, chat with document content using a local LLM (Gemma 7b via Ollama), and auto-generate flashcards for better learning. - Architected end-to-end document processing pipeline Integrating Marker for high-accuracy PDF-to-Markdown conversion, LangChain for text chunking, Sentence Transformers for embeddings, and ChromaDB as the vector database for semantic retrieval.	
Swipy (Mobile News Aggregator) <i>Flutter, Firebase, OpenAI API, REST APIs</i>	Sep 2025 – Nov 2025
- Developed a cross-platform news aggregator using Flutter with a swiping-style interface, integrating a third-party news API to fetch articles and an automated backend script that processes and stores curated content in Firebase. - Enhanced user engagement by generating concise article summaries on the fly using the OpenAI API, and implemented a personalized feed by filtering articles based on user-selected categories to surface relevant content.	

TECHNICAL SKILLS

Programming Languages: Python, C/C++, SQL, JavaScript/HTML/CSS
Frameworks & Libraries: PyTorch, LangChain, ChromaDB, Pandas, NumPy, Matplotlib, Flask, Django, React, Flutter, Firebase, REST APIs
Developer Tools: Git, VS Code, Jupyter
Languages: Arabic (Native), English (Fluent - TOEIC 970/990), French (Intermediate)